

Moment-Based Aggregation of Production Frontiers:

Second-Order Corrections, Their Failure at Capacity Thresholds,
and What Can Be Certified for Comparisons*

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Abstract

A population of production units shares a concave frontier f and differs in its inputs $x_i \in \mathbb{R}^d$. The aggregate output $Y = \sum_i f(x_i)$ is routinely replaced by the representative unit $Nf(\bar{x})$ or by its second-order correction $N[f(\bar{x}) + \frac{1}{2} \text{tr}(H_f(\bar{x})\Sigma)]$; we make the error of these approximations precise and test it in a controlled simulation. In the smooth regime, Taylor’s formula gives an exact remainder identity and a bound B_2 of the second-order error by the third absolute central moment of the inputs, computable for Cobb–Douglas frontiers; the improvement factor over the first order grows at least like the inverse dispersion (like its square for symmetric inputs), and the bounds hold in every simulated population. The improvement is certifiable from the input moments, their range and a derivative bound when $2B_2 < |Q_2|$, with Q_2 the second-order term; on log-normal inputs this holds for $\sigma \leq 0.042$, and a fourth-order certificate proved here, built from the signed third-moment tensor and a fourth moment, holds for $\sigma \leq 0.075$ in the same norm (fivefold improvement: $\sigma \leq 0.056$ instead of 0.018), against an observed improvement up to $\sigma = 0.75$ and a fivefold one up to $\sigma = 0.42$. At a capacity threshold $\min(f, c)$ the second-order error equals $-NT_u$ up to second-order terms, where T_u is the one-sided mean distance of the individual outputs to the capacity, linear in the dispersion when units straddle the threshold: the second order then improves nothing. Simulation confirms the predicted exponents (2, 4 and 4 on symmetric inputs; 1 at a crossing) and the crossing constant ($T_u/(\sigma\tau_z) = 0.997$ at $\sigma = 0.01$ on symmetric inputs); on log-normal inputs the second-order exponent lies between 3 and 4 and moves towards 4 as N grows (3.16 at $N = 200$, 3.96 at $N = 200000$), a finite- N effect of the sample third moment. In a two-level hierarchy the bottom-up second order beats the pooled one in every smooth cell (by factors 1.15 to 4473) and confines the failure at a capacity to the firms that straddle it, without removing it. A margin condition certifies the ordering of two aggregates (no reversal occurred among certified comparisons, as it guarantees), but covers a shrinking share of them and never exists at a crossing, where populations with identical first three moments can have aggregates that differ at first order. Two criteria were fixed in advance. A uniform fivefold error reduction (C1) holds for symmetric inputs and fails for log-normal inputs beyond $\sigma = 0.42$ and at crossings; a scalar recalibration of the second-order term does not restore it. A fivefold reduction of rank reversals in a decision experiment (C2) fails clearly in 14 of its 20 eligible capacity cells, with 1 capacity cell and 1 cell without capacity borderline at 1000 trials. Nothing is claimed about real populations or policy.

1 Introduction

Aggregating a nonlinear relation over heterogeneous units is an old problem with two classical faces. One asks whether an aggregate relation *exists* at all: for production functions the conditions are stringent [6, 7, 13], and for demand the representative-consumer conditions of Gorman [8] are equally special. The other asks, granting that one will use an aggregate relation anyway, how large the *aggregation bias* is and what it depends on [15, 16]. Moment-based expansions belong to the second face: the aggregate of a nonlinear micro function is expanded around the mean of the explanatory variables, and higher moments of their cross-sectional distribution enter as corrections [12, 17]. The zeroth term is the representative unit, the first correction is the Jensen gap [11], and the second-order approximation is the analogue of the second-order delta method [14]. A separate line, going back to Houthakker [10], obtains aggregate functional forms from the distribution of micro technologies; and the frontier literature [1, 5] describes units as lying on or below a common frontier, with idiosyncratic efficiency terms.

*Research line SGE001, recorded as “dynamic aggregation of production frontiers”. This draft contains no intertemporal model, so the word is not used in its title; see the paragraph on scope.

The research line to which this draft belongs asked, in the words of its public record, whether higher moments and second-order approximations in hierarchical production frontiers reduce aggregation errors, and how those errors affect comparisons and decisions; its recorded findings were that the approximations improve in smooth regimes but may fail at threshold crossings, and that a later calibration did not meet a predefined criterion. No manuscript survives; this draft reconstructs the line from that record: a precise mathematical core (Section 2), what can be proved about it (Section 3), a controlled simulation with two predefined criteria (Section 4, which states what is and is not documented about when they were fixed) and a list of every claim with its status (Section 5).

Scope. The frontier is a fixed, known, concave function; the inputs are drawn from stated distributions; nothing is estimated from data and nothing is claimed about any real sector or policy. The record of the line calls it “dynamic”; this draft studies the error along a sweep of the input dispersion and through the levels of a hierarchy (units, firms, sector) and introduces no intertemporal model, so the title does not use the word. The record also speaks of “hierarchical” frontiers; we read this as two-level aggregation (units, firms, sector) under a common frontier (Lemma 3.6, E3), not as frontiers that differ by level. Every measured number in the text is produced by the scripts of Appendix B with a fixed seed and injected as a macro.

2 Setting and definitions

Let $U \subseteq \mathbb{R}_{>0}^d$ be open and convex and $f: U \rightarrow \mathbb{R}$ a frontier, at least C^2 and, where stated, C^3 and concave. For a symmetric k -linear form T write $\|T\| = \sup_{\|h\|=1} |T[h, \dots, h]|$ (Euclidean norm on $\mathbb{R}^d$), so that $|T[h, \dots, h]| \leq \|T\| \|h\|^k$; for the Hessian this is the spectral norm, and for any k the Frobenius norm of the coefficient array dominates $\|T\|$ by Cauchy–Schwarz.

Definition 2.1 (Population, moments). A population is a finite list $x_1, \dots, x_N \in U$. Its mean is $\bar{x} = \frac{1}{N} \sum_i x_i$ (in U by convexity), its deviations are $h_i = x_i - \bar{x}$, and its central moments are

$$\Sigma = \frac{1}{N} \sum_i h_i h_i^\top, \quad \mu_3 = \frac{1}{N} \sum_i h_i \otimes h_i \otimes h_i, \quad s^2 = \text{tr } \Sigma = \frac{1}{N} \sum_i \|h_i\|^2, \quad m_3 = \frac{1}{N} \sum_i \|h_i\|^3.$$

Definition 2.2 (Aggregate and moment approximations). The aggregate output is $Y = \sum_{i=1}^N f(x_i)$. The approximations of order 1, 2, 3 are

$$\hat{Y}_1 = Nf(\bar{x}), \quad \hat{Y}_2 = \hat{Y}_1 + \frac{N}{2} \langle H_f(\bar{x}), \Sigma \rangle, \quad \hat{Y}_3 = \hat{Y}_2 + \frac{N}{6} \langle D^3 f(\bar{x}), \mu_3 \rangle,$$

where $\langle \cdot, \cdot \rangle$ is the entrywise inner product, so $\langle H_f, \Sigma \rangle = \text{tr}(H_f \Sigma)$. The *aggregation error of order k* is $E_k = Y - \hat{Y}_k$ and the relative error is $e_k = E_k/Y$. We write $Q_2 = \frac{N}{2} \text{tr}(H_f(\bar{x})\Sigma)$ for the second-order term, so $E_1 = E_2 + Q_2$.

$\hat{Y}_1$ is the representative unit; E_1/N is the Jensen gap of f on the empirical distribution of inputs, non-positive for concave f [11]. The approximations use only the moments of the inputs up to the stated order and the derivatives of f at the mean: this is what a statistician holding group means and covariances, but not the micro data, can compute.

Definition 2.3 (Hierarchy). Let the population be partitioned into groups (firms) $g = 1, \dots, G$ of sizes n_g , with means $\bar{x}_g$ and covariances Σ_g . The between-group and mean within-group covariances are $\Sigma_B = \frac{1}{N} \sum_g n_g (\bar{x}_g - \bar{x})(\bar{x}_g - \bar{x})^\top$ and $\bar{\Sigma}_W = \frac{1}{N} \sum_g n_g \Sigma_g$. Besides the *pooled* $\hat{Y}_2$ of Definition 2.2, the *bottom-up* estimator $\hat{Y}_2^{\text{BU}} = \sum_g n_g [f(\bar{x}_g) + \frac{1}{2} \text{tr}(H_f(\bar{x}_g)\Sigma_g)]$ applies the second order inside each firm and sums exactly, and the *top-down* estimator $\hat{Y}_2^{\text{TD}} = N[f(\bar{x}) + \frac{1}{2} \text{tr}(H_f(\bar{x})\Sigma_B)] + \frac{N}{2} \text{tr}(H_f(\bar{x})\bar{\Sigma}_W)$ expands the sum of firm outputs to second order in the firm means and adds the mean within-firm correction *evaluated at the sector mean*. So defined, it coincides with the pooled one (Lemma 3.6(i)): there are only two distinct second-order estimators, pooled and bottom-up, and E3 compares them.

Definition 2.4 (Capacity frontier, crossing term). For $c > 0$ the capacity frontier is $f_c = \min(f, c)$, concave whenever f is, and C^3 away from the level set $\{f = c\}$. Its moment approximations are those of Definition 2.2 with f_c in place of f : when $f(\bar{x}) > c$ the frontier is constant near $\bar{x}$ and $H_f[f_c](\bar{x}) = 0$, and when $f(\bar{x}) < c$ it equals $H_f(\bar{x})$. At $f(\bar{x}) = c$ the Hessian of f_c does not exist; “second-order approximation” there means, by convention, the one-sided derivatives of the “below capacity” branch (a convention about the estimator, not a property of f_c). Writing $u_i = f(x_i)$ for the uncapped individual outputs, $P_u = \sum_i (u_i - c)_+$, $Q_u = \sum_i (c - u_i)_+$, the *crossing term* is

$$T_u = \frac{1}{N} \min(P_u, Q_u),$$

the smaller of the mean output excess above capacity and the mean output shortfall below it. $T_u = 0$ exactly when all units lie on one side of the capacity.

3 Results

Throughout, $M_k^{(i)} = \sup_{t \in [0,1]} \|D^k f(\bar{x} + th_i)\|$ is the supremum of the k -th derivative along the segment from $\bar{x}$ to x_i , and $M_k = \max_i M_k^{(i)}$, which is at most the supremum over the convex hull of the population.

3.1 Smooth regime

Proposition 3.1 (Remainder identities and bounds). *Let $f \in C^2(U)$ and let $x_1, \dots, x_N \in U$.*

- (a) $E_1 = \frac{1}{2} \sum_i D^2 f(\xi_i)[h_i, h_i]$ for some ξ_i on the segment $[\bar{x}, x_i]$; hence $|E_1| \leq \frac{1}{2} \sum_i M_2^{(i)} \|h_i\|^2 =: B_1 \leq \frac{N}{2} M_2 s^2$, and $E_1 \leq 0$ if f is concave.
- (b) $E_2 = \sum_i \int_0^1 (1-t) (D^2 f(\bar{x} + th_i) - D^2 f(\bar{x}))[h_i, h_i] dt$; hence $|E_2| \leq \frac{1}{2} \sum_i \omega_i(\|h_i\|) \|h_i\|^2$, where $\omega_i(r) = \sup_{t \in [0,1]} \|D^2 f(\bar{x} + th_i) - D^2 f(\bar{x})\|$ is a modulus of continuity of the Hessian along the segment.
- (c) If $f \in C^3(U)$ then $E_2 = \sum_i \int_0^1 \frac{(1-t)^2}{2} D^3 f(\bar{x} + th_i)[h_i, h_i, h_i] dt$ and $|E_2| \leq \frac{1}{6} \sum_i M_3^{(i)} \|h_i\|^3 =: B_2 \leq \frac{N}{6} M_3 m_3$.
- (d) If moreover $D^2 f \preceq -\lambda I$ on the convex hull of the population for some $\lambda > 0$, then $|E_1| \geq \frac{\lambda}{2} N s^2$ and $\frac{|E_2|}{|E_1|} \leq \frac{M_3 m_3}{3\lambda s^2} \leq \frac{M_3 \rho}{3\lambda}$, $\rho = \max_i \|h_i\|$.
- (e) (Certificates; under the hypothesis of (c).) If $(k+1)B_2 < |Q_2|$ for some $k \geq 1$, then $|E_1| \geq |Q_2| - B_2 > kB_2 \geq k|E_2|$: $k=1$ certifies that the second order improves on the first, $k=5$ the fivefold improvement of criterion C1 (Section 4). With the moment form $B_2 \leq \frac{N}{6} M_3 m_3$, where M_3 bounds $\|D^3 f\|$ on a known region containing the population (for instance the coordinate box of the input range), the certificate uses only $\bar{x}$, Σ , m_3 and that region: the moment-and-range certificate. With the segmentwise B_2 it needs the individual inputs, though not f at the units: the micro-data certificate, a diagnostic, since whoever holds the inputs and f can compute Y . As $B_2 = O(\sigma^3)$ and $|Q_2| = \Theta(\sigma^2)$ under (d), both hold for small dispersion.

Proof. Taylor's formula along the segment $t \mapsto \bar{x} + th_i$, which lies in U [4, Ch. VIII], gives the Lagrange form of order two and the integral forms of orders two and three for $f(x_i) - f(\bar{x}) - Df(\bar{x})[h_i]$. Summing over i , the linear terms vanish because $\sum_i h_i = 0$ and the quadratic terms add up to $\frac{N}{2} \text{tr}(H_f(\bar{x})\Sigma) = Q_2$; this gives (a) and (c), and (b) after writing $\frac{1}{2} D^2 f(\bar{x})[h_i, h_i] = \int_0^1 (1-t) D^2 f(\bar{x})[h_i, h_i] dt$. The bounds use $|T[h, \dots, h]| \leq \|T\| \|h\|^k$, $\int_0^1 (1-t) dt = \frac{1}{2}$, $\int_0^1 \frac{(1-t)^2}{2} dt = \frac{1}{6}$ and $\|h_i\|^3 \leq \rho \|h_i\|^2$. Concavity gives $D^2 f(\xi_i)[h_i, h_i] \leq 0$ in (a), and $D^2 f \preceq -\lambda I$ gives $-D^2 f(\xi_i)[h_i, h_i] \geq \lambda \|h_i\|^2$ in (d). For (e), $|E_1| = |E_2 + Q_2| \geq |Q_2| - |E_2|$ and $|E_2| \leq B_2$. $\square$

Remark 3.2 (Orders in the dispersion; constants). Fix a centred configuration $z_1, \dots, z_N$ ($\sum z_i = 0$) and let $x_i = \bar{x} + \sigma z_i$. Then $s^2 = \sigma^2 s_z^2$, $m_3 = \sigma^3 m_z^3$, and Proposition 3.1 gives $|E_1| = \Theta(\sigma^2)$ under (d), $|E_2| = O(\sigma^3)$ and $|E_2|/|E_1| = O(\sigma)$: the improvement factor of the second order over the first grows at least like $1/\sigma$. If $D^3 f$ is Lipschitz on the convex hull (for instance $f \in C^4$, as Cobb–Douglas and CES are), the identity in (c) gives $E_2 = \frac{\sigma^3}{6} N \langle D^3 f(\bar{x}), \mu_3^z \rangle + O(\sigma^4)$: for a configuration with vanishing third central moments $E_2 = O(\sigma^4)$, the improvement factor grows like σ^{-2} and $\hat{Y}_3 = \hat{Y}_2$, and the third-order correction is useful only when the third central moments dominate the fourth-order remainder. For inputs $\bar{x} \odot \exp(\sigma z - \sigma^2/2)$ with Gaussian z two facts follow. The *population* third central moments are $O(\sigma^4)$ (the skewness of a log-normal variable is $\approx 3\sigma$ for small σ), the same order as the neglected term, so no improvement of the third over the second order is predicted. But the configuration is a finite sample of N units: its *sample* third central moment is the population one plus a sampling fluctuation of order $\sigma^3 N^{-1/2}$, which does not vanish at fixed N ; hence at fixed N and $\sigma \rightarrow 0$ the exponent of E_2 in σ is 3, and 4 is approached only for $\sigma \gg N^{-1/2}$. E_1 measures both effects. The orders follow from (c); the log-normal moment orders are standard calculations, given without proof. For Cobb–Douglas frontiers the suprema $M_k^{(i)}$ have closed forms (Lemma 3.4).

Proposition 3.3 (Fourth-order certificate). *Let $f \in C^4(U)$, $x_1, \dots, x_N \in U$, and let $C_3 = \frac{N}{6} \langle D^3 f(\bar{x}), \mu_3 \rangle = \hat{Y}_3 - \hat{Y}_2$, built from the signed third central moment tensor. Fix an invertible linear map S of $\mathbb{R}^d$ (below $S = I$ or $S = \text{diag}(\bar{x})$), write $\|T\|_S = \|T[S \cdot, \dots, S \cdot]\|$ for a symmetric 4-linear form T , $M_{4,S}^{(i)} = \sup_{t \in [0,1]} \|D^4 f(\bar{x} + th_i)\|_S$ and $m_4^S = \frac{1}{N} \sum_i \|S^{-1} h_i\|^4$.*

- (a) $E_2 = C_3 + E_3$ with $E_3 = Y - \hat{Y}_3 = \sum_i \int_0^1 \frac{(1-t)^3}{6} D^4 f(\bar{x} + th_i) [h_i, h_i, h_i, h_i] dt$.
- (b) $|E_3| \leq \frac{1}{24} \sum_i M_{4,S}^{(i)} \|S^{-1} h_i\|^4 =: B_3 \leq \frac{N}{24} M_{4,S} m_4^S$ whenever $M_{4,S} \geq \max_i M_{4,S}^{(i)}$, for instance $M_{4,S} = \sup_R \|D^4 f\|_S$ over a convex region $R \subseteq U$ containing the population. Hence $|E_2| \leq \min(B_2, |C_3| + B_3)$.
- (c) (Certificate.) $|E_1| \geq |Q_2 + C_3| - B_3 \geq |Q_2| - |C_3| - B_3$. If $(k+1)(|C_3| + B_3) < |Q_2|$ for some $k \geq 1$, or more generally $|Q_2 + C_3| - B_3 > k(|C_3| + B_3)$, then $|E_1| > k(|C_3| + B_3) \geq k|E_2|$.
- (d) (Comparisons.) $|(Y^A - Y^B) - (\hat{Y}_3^A - \hat{Y}_3^B)| \leq B_3^A + B_3^B$ for two populations A, B ; and if $|\hat{Y}_2^A - \hat{Y}_2^B| > (|C_3^A| + B_3^A) + (|C_3^B| + B_3^B)$, then $\hat{Y}_2^A - \hat{Y}_2^B, \hat{Y}_3^A - \hat{Y}_3^B$ and $Y^A - Y^B$ have the same sign.

With the moment form of (b) the certificate uses $\bar{x}$, Σ , the whole tensor μ_3 , m_4^S and the region R (fourth-order moment-and-range certificate; unlike m_3 , $m_4^S = \sum_{j,l} (\mu_4^S)_{jjll}$ is a polynomial moment); with the segmentwise B_3 it needs the individual inputs (micro-data certificate). Proofs of Proposition 3.3 and Lemma 3.4 are in Appendix A.

Lemma 3.4 (Derivatives of a Cobb–Douglas frontier). *Let $f(x) = A \prod_{m=1}^d x_m^{a_m}$ with $A > 0$, $0 < a_m < 1$. For multiplicities $e = (e_1, \dots, e_d)$, $\sum_m e_m = k$, $\partial^e f(x) = A \prod_m (a_m)_{e_m} x_m^{a_m - e_m}$ with the falling factorial $(a)_n = a(a-1)\dots(a-n+1)$. On a box $[\ell, u] \subset \mathbb{R}_{>0}^d$, $|\partial^e f|$ is largest at the corner c^e with $c_m^e = u_m$ if $e_m = 0$ and ℓ_m otherwise, and for $S = \text{diag}(s)$, $s > 0$, $\sup_{[\ell, u]} \|D^k f\|_S \leq (\sum_{j_1, \dots, j_k} (s^e |\partial^e f(c^e)|)^2)^{1/2}$, where $s^e = \prod_m s_m^{e_m}$ and e are the multiplicities of $(j_1, \dots, j_k)$.*

Remark 3.5 (What changes in the order, and what does not). With vanishing third central moments, $B_2 = \Theta(\sigma^3)$ but $|E_2|$ and $|C_3| + B_3$ are $O(\sigma^4)$ (Remark 3.2). For log-normal inputs, in the coordinates scaled by $\bar{x}$ and at leading order (population moments, derivatives at $\bar{x}$), $B_2/|Q_2| \approx 2.39\sigma$, $(|C_3| + B_3)/|Q_2| \approx 5.06\sigma^2$ and $|E_2|/|Q_2| \approx 0.70\sigma^2$, so the certified dispersion scales like $1/(k+1)$ for the third-order certificate and like $1/\sqrt{k+1}$ for the fourth-order one: the gain is larger for $k=5$. At finite N the suprema are taken over the input range, which for log-normal inputs grows like $\bar{x} \odot e^{\pm \sigma z_{\max}}$ with $z_{\max} \approx \sqrt{2 \log N}$, and $\|D^k f\|$ grows at the lower corner roughly like $e^{(k-a_m)\sigma z_{\max}}$, faster for $k=4$. What remains is the norm inequality times this range inflation, not an order; as the inflation is stronger for $D^4 f$, B_2 can be the smaller bound at large dispersion (hence the minimum in (b)). These orders are heuristic; the certificate is exact.

3.2 Hierarchy

Lemma 3.6 (Two levels). *With the notation of Definition 2.3:*

- (i) $\Sigma = \bar{\Sigma}_W + \Sigma_B$ (law of total covariance), and consequently $\hat{Y}_2^{\text{TD}} = \hat{Y}_2$.
- (ii) $Y - \hat{Y}_2^{\text{BU}} = \sum_g E_2^{(g)}$, the sum of the within-firm second-order errors, so $|Y - \hat{Y}_2^{\text{BU}}| \leq \sum_g B_2^{(g)}$ with the firm-level bounds of Proposition 3.1(c), which involve only within-firm deviations.
- (iii) If f is a polynomial of degree at most 2 then $\hat{Y}_2 = \hat{Y}_2^{\text{BU}} = Y$.

Proof. (i) $h_i = (x_i - \bar{x}_g) + (\bar{x}_g - \bar{x})$ for i in group g and the cross terms vanish on summing within the group; then $\hat{Y}_2^{\text{TD}} = N[f(\bar{x}) + \frac{1}{2} \text{tr}(H_f(\bar{x})(\Sigma_B + \bar{\Sigma}_W))] = \hat{Y}_2$. (ii) is the definition of $\hat{Y}_2^{\text{BU}}$ with $Y = \sum_g \sum_{i \in g} f(x_i)$. (iii) The remainders in Proposition 3.1(c) vanish when $D^3 f \equiv 0$. $\square$

The lemma is elementary; it says that hierarchy buys exactly one thing, evaluating f and its Hessian at the firm means instead of the sector mean, whose price in the pooled error is the between-firm third-order term. Bottom-up aggregation isolates the error firm by firm; in particular (Section 3.3) only the firms whose units straddle a capacity contribute a first-order-in-dispersion error.

3.3 Threshold crossings

Lemma 3.7 (Hinge identity). *For reals $u_1, \dots, u_N$ with mean $\bar{u}$ and any c , with P_u, Q_u as in Definition 2.4, $\sum_i (u_i - c)_+ - N(\bar{u} - c)_+ = \min(P_u, Q_u)$.*

Proof. $P_u - Q_u = \sum_i (u_i - c) = N(\bar{u} - c)$, so $N(\bar{u} - c)_+ = (P_u - Q_u)_+$ and $P_u - (P_u - Q_u)_+ = \min(P_u, Q_u)$. $\square$

Proposition 3.8 (Second-order error at a capacity). *Let $f \in C^3(U)$, $c > 0$, and let $E_k(f)$, $E_k(f_c)$, Q_2 , B_1 , B_2 refer to the same population. Then*

$$E_2(f_c) = -NT_u + E_2(f) + Q_2 \mathbf{1}\{f(\bar{x}) > c\} - N[(\bar{u} - c)_+ - (f(\bar{x}) - c)_+], \quad (1)$$

where $\bar{u} = \frac{1}{N} \sum_i u_i = Y(f)/N$. Consequently

$$|E_2(f_c) + NT_u| \leq B_2 + B_1 + |Q_2| \mathbf{1}\{f(\bar{x}) > c\}, \quad E_1(f_c) = E_2(f_c) + Q_2 \mathbf{1}\{f(\bar{x}) \leq c\}. \quad (2)$$

Proof. $f_c = f - (f - c)_+$, so $Y(f_c) = Y(f) - P_u$. By Definition 2.4, $\hat{Y}_2(f_c) = Nf(\bar{x}) - N(f(\bar{x}) - c)_+ + Q_2 \mathbf{1}\{f(\bar{x}) \leq c\}$. Subtracting, $E_2(f_c) = [Y(f) - Nf(\bar{x}) - Q_2] + Q_2 \mathbf{1}\{f(\bar{x}) > c\} - [P_u - N(f(\bar{x}) - c)_+]$, and the first bracket is $E_2(f)$. By Lemma 3.7, $P_u - N(f(\bar{x}) - c)_+ = \min(P_u, Q_u) + N[(\bar{u} - c)_+ - (f(\bar{x}) - c)_+]$, which gives (1). For (2): the hinge is 1-Lipschitz, so the last term of (1) is at most $N|\bar{u} - f(\bar{x})| = |E_1(f)| \leq B_1$, and $|E_2(f)| \leq B_2$ by Proposition 3.1(c). The relation between $E_1(f_c)$ and $E_2(f_c)$ is the definition of the second-order term of f_c . $\square$

Corollary 3.9 (Failure at a crossing). *Let $x_i = \bar{x} + \sigma z_i$ with a fixed centred configuration (z_i) , let $f(\bar{x}) = c$, $g = \nabla f(\bar{x})$ and $\tau_z = \frac{1}{2N} \sum_i |g \cdot z_i|$. Then $|T_u - \sigma \tau_z| \leq \frac{M_2}{2} \sigma^2 s_z^2$ and*

$$|E_2(f_c) + N\sigma \tau_z| \leq N \left[M_2 \sigma^2 s_z^2 + \frac{M_3}{6} \sigma^3 m_3^z \right].$$

If $\tau_z > 0$ (some z_i is not orthogonal to g , that is, the linearised outputs straddle the capacity), then as $\sigma \rightarrow 0$: $E_2(f_c) = -N\sigma \tau_z(1 + o(1))$, so the relative second-order error is of order σ ; $|E_1(f_c)|/|E_2(f_c)| \rightarrow 1$; and $|E_2(f_c)|/|E_2(f)| \geq 3\tau_z/(M_3 m_3^z \sigma^2) \rightarrow \infty$.

Proof. By Taylor's formula $u_i - c = \sigma g \cdot z_i + r_i$ with $|r_i| \leq \frac{M_2}{2} \sigma^2 \|z_i\|^2$. The hinge and min are 1-Lipschitz in each argument, so $|P_u - \sigma \sum_i (g \cdot z_i)_+|$ and $|Q_u - \sigma \sum_i (g \cdot z_i)_-|$ are at most $\frac{M_2}{2} \sigma^2 N s_z^2$, and so is $|NT_u - \sigma \min(\sum (g \cdot z_i)_+, \sum (g \cdot z_i)_-)|$. Since $\sum_i g \cdot z_i = 0$, both sums equal $\frac{1}{2} \sum_i |g \cdot z_i| = N\tau_z$. The displayed bound follows from (2) with the indicator equal to 0 (the convention at $f(\bar{x}) = c$), $B_1 \leq \frac{N}{2} M_2 \sigma^2 s_z^2$ and $B_2 \leq \frac{N}{6} M_3 \sigma^3 m_3^z$. The limits follow: $E_1(f_c) - E_2(f_c) = Q_2 = O(N\sigma^2)$, and $|E_2(f)| \leq \frac{N}{6} M_3 \sigma^3 m_3^z$ while $|E_2(f_c)| \geq N\sigma \tau_z/2$ for small σ . $\square$

Remark 3.10 (Hypotheses, shifted mean, uniformity in N). (i) If the mean sits at output distance σb from the capacity, $f(\bar{x}) = c + \sigma b$, the same argument gives $T_u = \sigma \tau_z(b) + O(\sigma^2)$ with $\tau_z(b) = \min(\frac{1}{N} \sum (g \cdot z_i + b)_+, \frac{1}{N} \sum (g \cdot z_i + b)_-)$, which is positive exactly when $-b$ lies strictly inside the range of the $g \cdot z_i$: the order- σ error appears when, and only when, units lie on both sides of the capacity; when all units are on one side, $T_u = 0$ and Proposition 3.1 applies to the smooth branch. (ii) The corollary applies verbatim to the symmetric design of Section 4 (antithetic pairs: the sample mean is exactly $\bar{x}$ and $z_i = \bar{x} \odot w_i$ is fixed along the sweep). It does not apply verbatim to the log-normal design, where the centred configuration $(x_i - \bar{x})/\sigma$ depends on σ and the sample mean satisfies $f(\bar{x}) - c = O(\sigma N^{-1/2})$, so that (i) with a random b of order $N^{-1/2}$ is the applicable statement; E2 checks both. (iii) The constants depend on the population only through s_z^2 , m_3^z and the derivative suprema M_2, M_3 over its convex hull; the relative error bounds are therefore uniform in N whenever these are bounded uniformly in N , which is not automatic (for Gaussian z the hull grows like $\sqrt{\log N}$, and the Cobb–Douglas derivatives diverge at the boundary of $\mathbb{R}_{>0}^d$).

Example 3.11 (Moments do not determine the aggregate at a crossing). Take $d = 1$, $f(x) = x$, $f_c = \min(x, c)$, and the two populations $\{c - \sigma, c - \sigma, c + \sigma, c + \sigma\}$ and $\{c - \sigma\sqrt{2}, c, c, c + \sigma\sqrt{2}\}$. Both have mean c , variance σ^2 and third central moment 0 (they differ in the fourth, σ^4 against $2\sigma^4$), so every approximation built from $(\bar{x}, \Sigma, \mu_3)$ takes the same value on both; but their aggregates are $4c - 2\sigma$ and $4c - \sqrt{2}\sigma$ ($Y = Nc - NT_u$ here). Any estimator that depends on the population only through its first three moments has error at least $(1 - \frac{\sqrt{2}}{2})\sigma$ on one of them: at a crossing the missing information is of first order in the dispersion, and no recalibration of a second-moment term can supply it uniformly. E5 below tests this on the simulated populations.

3.4 Comparisons

Proposition 3.12 (Margin condition). *Let Y^A, Y^B be two aggregates with approximations $\hat{Y}^A, \hat{Y}^B$ and certified bounds $|Y^A - \hat{Y}^A| \leq B^A$, $|Y^B - \hat{Y}^B| \leq B^B$. If $|\hat{Y}^A - \hat{Y}^B| > B^A + B^B$ then $\text{sign}(Y^A - Y^B) = \text{sign}(\hat{Y}^A - \hat{Y}^B)$.*

Proof. $Y^A - Y^B = (\hat{Y}^A - \hat{Y}^B) + (E^A - E^B)$ and $|E^A - E^B| \leq B^A + B^B < |\hat{Y}^A - \hat{Y}^B|$. $\square$

In the smooth regime B^A, B^B can be the bounds of Proposition 3.1(c) or 3.3(b): in moment form a comparison is then *certified* from the moments, the input range and a derivative bound, in segmentwise form from the individual inputs. At a crossing the error is $-NT_u$ up to smaller terms and T_u is not a function of the moments (Example 3.11), so no certificate of this kind exists. A closed-form example shows that the condition is not idle.

Example 3.13 (A reversal corrected at second order). $f(x) = \sqrt{x}$; sector A has all units at 1, sector B half at $1 + \delta + \sigma$ and half at $1 + \delta - \sigma$. With $\delta = 0.01$, $\sigma = 0.3$: $Y^B/N = 0.9936 < 1 = Y^A/N$, while $\hat{Y}_1^B/N = \sqrt{1 + \delta} = 1.0050 > 1$ (reversed) and $\hat{Y}_2^B/N = \sqrt{1 + \delta} - \frac{1}{8}(1 + \delta)^{-3/2}\sigma^2 = 0.9939 < 1$ (correct). On a capped frontier no order helps: the two populations of Example 3.11 cannot be ordered by any approximation built from their first three moments.

frontier	law	σ	$ e_1 $	$ e_2 $	$ e_3 $	min $ E_1 / E_2 $	med $ E_2 / E_3 $	B_2/Y
CD	LN	0.01	1.6×10^{-5}	2.6×10^{-9}	1.5×10^{-9}	2366.9	1.81	1.1×10^{-6}
CD	LN	0.1	1.6×10^{-3}	9.7×10^{-6}	1.5×10^{-5}	94.7	0.65	1.3×10^{-3}
CD	LN	1	1.7×10^{-1}	1.6×10^{-1}	9.4×10^{-1}	0.3	0.17	6.0×10^1
CD	SU	0.01	1.6×10^{-5}	8.3×10^{-10}	8.3×10^{-10}	18407.6	1.00	9.0×10^{-7}
CD	SU	0.0935	1.4×10^{-3}	6.4×10^{-6}	6.4×10^{-6}	209.4	1.00	8.5×10^{-4}
CD	SU	0.5	5.0×10^{-2}	8.1×10^{-3}	8.1×10^{-3}	5.9	1.00	1.6×10^0

Table 1: E1, Cobb–Douglas frontier. Median relative errors over 20 replications, $N = 2000$, at three of the sweep’s dispersions; the improvement ratio of the second over the first order is the minimum over replications; B_2/Y is the median certified bound of Proposition 3.1(c). The CES rows and the full sweep, with bootstrap intervals, are in `results/tables.md`.

4 Experiments

Protocol. Two inputs (capital, labour) with sector mean $\bar{x} = (2.0, 1.0)$. Two smooth frontiers: Cobb–Douglas $f = K^{0.3}L^{0.5}$ [3] and CES $f = (0.4K^{-1} + 0.6L^{-1})^{-0.8}$ [2], both with returns to scale 0.8 and hence strictly concave on $\mathbb{R}_{>0}^2$; their analytic gradient, Hessian and third-derivative tensor are checked against central differences (E0: worst relative deviation 1.7×10^{-10} and 5.7×10^{-10}). Two input laws with correlation 0.5 between the standard shocks: log-normal, $x = \bar{x} \odot \exp(\sigma z - \sigma^2/2)$ (LN, skewed), and symmetric uniform, $x = \bar{x} \odot (1 + \sigma w)$ with w uniform on $[-\sqrt{3}, \sqrt{3}]$ through a Gaussian copula and antithetic pairs $(w, -w)$, so that the sample mean is exactly $\bar{x}$ and all sample third central moments vanish exactly (SU). Shocks are drawn once per replication and reused across the dispersion sweep (common random numbers); each experiment has its own generator spawned from the master seed. Errors are reported as medians over replications of $|e_k|$, with percentile bootstrap intervals over replications in `results/tables.md`; ratios of errors are the minimum over replications, the conservative choice for a uniform criterion. Bounds are computed for the Cobb–Douglas frontier (for CES see the Limitations), in segmentwise (micro-data) and in moment-and-range form (Proposition 3.1(e), Appendix B).

Predefined criteria. *C1 (certified improvement):* for a given frontier and input law, the minimum over replications of $|E_1|/|E_2|$ is at least 5 at every σ of the sweep (log-spaced in $[0.01, 1]$ for LN and in $[0.01, 0.5]$ for SU). *C2 (decisions):* in every cell of the decision experiment in which the first-order reversal rate is at least 2%, the second-order reversal rate is at most one fifth of it. Both criteria were written into the script before the first run (v0.1); no dated record of that version exists, so from v0.2 on the script’s SHA-256 (5b309162009c) and the run date (2026-10-03) are stored with the results.

4.1 E1: dispersion sweep on smooth frontiers

$N = 2000$ units, 20 replications. Figure 1 and Table 1. The small-dispersion exponents (least squares on the median curves for $\sigma \leq 0.05$; bootstrap standard errors over replications in parentheses when they exceed 0.005) are: order 1, 2.00 (CD–LN), 2.00 (CD–SU), 2.00 and 2.00 (CES); order 2, 3.12 (0.26), 4.00, 3.18 (0.24), 4.00; order 3, 4.01, 4.00, 4.00, 4.00. These are the exponents of Remark 3.2: 2 for the first order, 4 for the second order on symmetric inputs and for the third order on both laws. On log-normal inputs the second-order exponent lies between 3 and 4 because the sample third central moment of a finite population does not vanish: its $O(N^{-1/2})$ sampling fluctuation contributes a σ^3 term that dominates the $O(\sigma^4)$ population skewness for $\sigma \lesssim N^{-1/2}$, which is inside the fitting window at $N = 2000$. A dedicated run (E1c: Cobb–Douglas, LN, 60 replications for each $N \in \{200, 2000, 20000, 200000\}$, bootstrap standard errors) gives second-order exponents 3.16 (0.09), 3.50 (0.10), 4.05 (0.10) and 3.96 (0.02), and third-order exponents 4.00, 4.00, 4.00, 4.00: the third order removes the sample third moment exactly, whatever N . The third-moment term $|E_2 - E_3|/Y$ at $\sigma = 0.01$ has median 8.0×10^{-9} , 2.3×10^{-9} , 2.3×10^{-9} and 2.6×10^{-9} : a part that decreases with N on top of a floor set by the population skewness, as Remark 3.2 says.

Criterion C1. On symmetric inputs C1 holds on the whole sweep for both frontiers (minimum ratios 5.87 and 5.64 at $\sigma = 0.5$). On log-normal inputs it holds at $\sigma = 0.42$ and fails at the next grid point, $\sigma = 0.56$ (both frontiers); at $\sigma = 1.0$ the minimum ratio is 0.31 (CD) and 0.48 (CES), that is, the second-order approximation is then *worse* than the representative unit.

Third order. On log-normal inputs the median ratio $|E_2|/|E_3|$ is below 1 in 15 of 17 cells for Cobb–Douglas and 13 of 17 for CES: adding the third moment does not improve the aggregate, as Remark 3.2 anticipates, and at large dispersion it degrades it substantially (minimum ratio 0.002). On symmetric inputs $\hat{Y}_3 = \hat{Y}_2$ exactly.

Bounds and certificates. The segmentwise bound B_2 holds in every replication and cell of both input laws

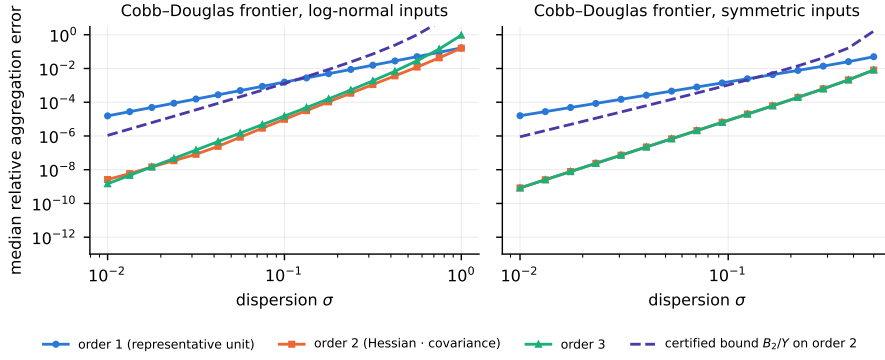


Figure 1: E1, Cobb–Douglas frontier (the CES curves are the same up to constants). Median relative aggregation error against dispersion for the three orders and the certified bound B_2/Y on the second-order error. Slopes 2, 4 (3–4 on skewed inputs at this N) and 4.

law	σ	$ e_1 $	$ e_2 $	$ e_2 $ (smooth)	$\min \frac{ E_1 }{ E_2 }$	below branch	above (%)	$\frac{-E_2}{NT_u}$	$\frac{T_u}{\sigma\tau_z}$	$\left \frac{T_u}{\sigma\tau_z(b)} - 1 \right $
LN	0.01	2.8×10^{-3}	2.8×10^{-3}	2.4×10^{-9}	1.000	1.005	50	1.000	0.993	0.0028
LN	0.0316	8.9×10^{-3}	8.8×10^{-3}	1.2×10^{-7}	1.000	1.017	50	1.000	0.989	0.0087
LN	0.1	2.9×10^{-2}	2.8×10^{-2}	1.1×10^{-5}	1.000	1.056	50	1.006	0.973	0.0243
LN	0.316	1.0×10^{-1}	1.0×10^{-1}	1.1×10^{-3}	1.000	1.202	55	1.150	0.909	0.0901
SU	0.01	3.0×10^{-3}	3.0×10^{-3}	8.3×10^{-10}	1.005	1.005	0	1.000	0.997	0.0027
SU	0.0316	9.5×10^{-3}	9.4×10^{-3}	8.3×10^{-8}	1.016	1.016	0	1.000	0.992	0.0083
SU	0.1	3.1×10^{-2}	3.0×10^{-2}	8.4×10^{-6}	1.051	1.051	0	1.000	0.975	0.0253
SU	0.316	1.2×10^{-1}	9.8×10^{-2}	9.6×10^{-4}	1.167	1.167	0	1.011	0.929	0.0712

Table 2: E2 at $\delta = 0$ (mean output on the capacity). “Smooth” is the error of the same population on the uncapped frontier; the improvement ratio is the minimum over all replications and over the replications whose sample mean falls below the capacity (“below branch”); “above” is the percentage of replications whose sample mean falls above it (SU: fixed to the below branch by convention); the last three columns are medians over replications of $-E_2/(NT_u)$ (Proposition 3.8), of $T_u/(\sigma\tau_z)$ (Corollary 3.9) and of the absolute deviation from 1 of the shifted ratio of Remark 3.10(i).

(checks: yes, yes), as does the moment-and-range bound $\frac{N}{6}M_3m_3$ with M_3 taken over the coordinate box of the population, which is larger by a factor that reaches 7215 at the largest dispersion. Both are conservative: the largest observed $|E_2|/B_2$ is 0.018 (LN) and 0.016 (SU), because the actual error cancels to order σ^3 – σ^4 while the bound is built from the absolute third moment. The certificate $2B_2 < |Q_2|$ of Proposition 3.1(e) holds in every replication for $\sigma \leq 0.042$ (LN) and $\sigma \leq 0.053$ (SU) in its moment-and-range form, and for $\sigma \leq 0.056$ and $\sigma \leq 0.071$ in its micro-data form; the fivefold improvement of C1 ($6B_2 < |Q_2|$) only for $\sigma \leq 0.018$ and 0.023 (micro data: 0.018, 0.023). The improvement certified, $|E_2| < |E_1|$, is observed in every replication up to $\sigma = 0.75$ (LN, both frontiers) and on the whole SU sweep, the fivefold one up to 0.42 (LN).

Fourth-order certificate. The bounds $|E_3| \leq B_3$ and $|E_2| \leq |C_3| + B_3$ of Proposition 3.3 hold in every replication and cell of both laws and both frontiers (all hold: yes), and even in moment-and-range form $|E_2|/(|C_3| + B_3)$ reaches 0.18 (LN) and 0.054 (SU), against at most 0.018 and 0.016 for the micro-data B_2 . In the same Euclidean norm, $2(|C_3| + B_3) < |Q_2|$ holds in every replication for $\sigma \leq 0.075$ (LN) and 0.124 (SU) from moments and range and for $\sigma \leq 0.100$ and 0.164 from micro data, and the fivefold improvement is certified up to 0.056 and 0.071 (micro data: 0.075, 0.094). A change of norm also helps, and we keep the two gains apart: with deviations measured relative to the mean ($S = \text{diag}(\bar{x})$) the *third*-order moment-and-range certificate (LN) already reaches 0.075 ($k = 1$) and 0.032 ($k = 5$), and at that norm the fourth order takes it to 0.100 and 0.075 (micro data: from 0.133 and 0.056 to 0.178 and 0.133). These dispersions are grid points (adjacent ones differ by a factor $10^{1/8}$), so each gain is resolved to one step. The signed form of Proposition 3.3(c) certifies the same dispersions (all variants: yes; on SU inputs $C_3 = 0$). As Remark 3.5 predicts, the gain is larger for $k = 5$; the certified range stays far below the observed $\sigma = 0.75$ and 0.42 (LN).

4.2 E2: capacity thresholds

Cobb–Douglas frontier, $N = 2000$, 20 replications, both input laws, capacity $c = f(\bar{x})/(1 + \delta)$ with $\delta \in \{-0.30, -0.10, -0.03, +0.00, +0.03, +0.10, +0.30\}$ (negative: the representative unit is below capacity) and σ from 0.01 to 1 (0.5 for SU). Figure 2 (top) and Table 2; the full grid is in `results/tables.md`.

With the mean on the capacity ($\delta = 0$) the small-dispersion exponent of the second-order error is 1.01 (LN; bootstrap s.e. 0.003) and 1.00 (SU), against 3.52 (0.21) and 4.00 for the same populations on the uncapped

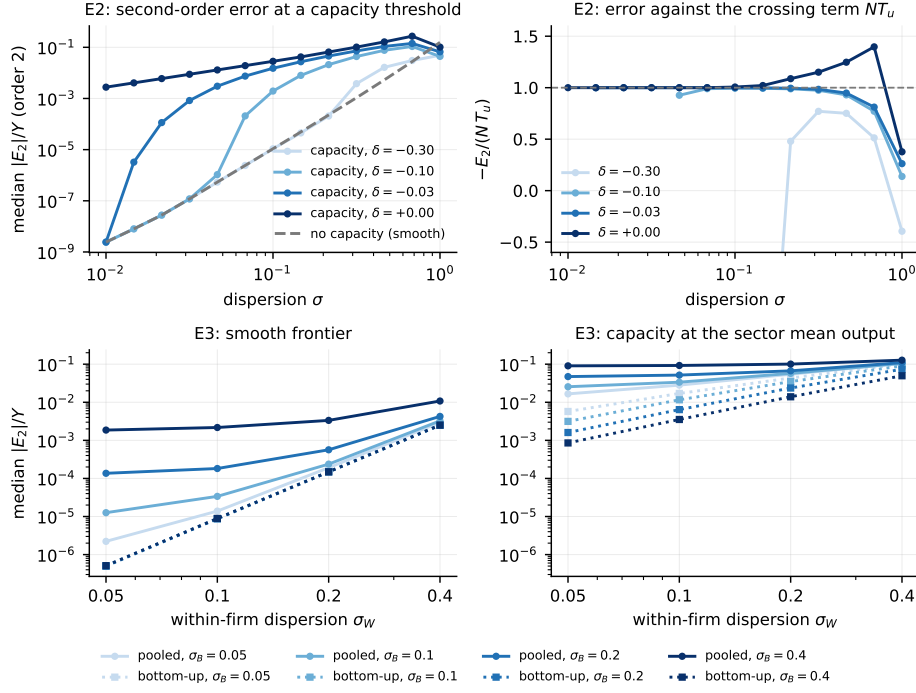


Figure 2: Top (E2, log-normal inputs): second-order relative error against dispersion for the capped frontier at four threshold proximities and for the uncapped frontier (left); the ratio $-E_2/(NT_u)$, equal to $1 + O(\sigma)$ by Proposition 3.8 (right; one point of $\delta = -0.30$ at $\sigma \approx 0.15$, where NT_u is smaller than the smooth error and the ratio is -3.2 , lies below the axis). Bottom (E3): pooled (solid) and bottom-up (dotted) second-order errors against within-firm dispersion, one shade per between-firm dispersion, for the smooth frontier (left) and with a capacity at the sector mean output (right).

frontier: the error is linear in the dispersion, as Corollary 3.9 states, and at $\sigma = 0.01$ it is 1×10^6 times the uncapped one (LN). The constant of the corollary is checked on the symmetric design, to which it applies verbatim: the median of $T_u/(\sigma\tau_z)$ is 0.997 at $\sigma = 0.01$ (bootstrap interval $[0.997, 0.997]$), 0.975 at $\sigma = 0.1$ and 0.929 at $\sigma = 0.316$, a deviation linear in σ . On the log-normal design the sample mean is off the capacity ($|f(\bar{x}) - c|/c$ up to 2.9×10^{-4} at $\sigma = 0.01$) and the shifted ratio $T_u/(\sigma\tau_z(b))$ of Remark 3.10(i) applies: at $\sigma = 0.01$ it lies in $[0.997, 1.003]$ in every replication, with the sign of the deviation set by the side of the capacity on which the sample mean falls ($P_u < Q_u$ in 50% of the replications, so the median falls between two groups and says nothing), and its median absolute deviation from 1 is 0.0028, the same $O(\sigma)$ deviation as on symmetric inputs (0.0027). Identity (1) holds to a relative 2×10^{-15} in all 168 cells (a check of the code), so $-E_2/(NT_u)$ is not an independent test of the constant (it equals $1 - E_2(f)/(NT_u)$ in the below branch and 1 in the above branch with $\bar{u} > c$); the two-sided bound (2) holds in all cells because it reduces to $B_1 \geq |E_1(f)|$ and $B_2 \geq |E_2(f)|$, and its lower-bound form is informative where NT_u exceeds the smooth bounds ($\sigma \leq 0.15$ at $\delta = 0$, LN; 33 of the 168 cells).

At $\delta = 0$ the LN sample mean falls above the capacity in 40–55% of the replications, where $\hat{Y}_2 = \hat{Y}_1$ by construction and the improvement ratio is exactly 1; SU is fixed to the below branch by the convention of Definition 2.4 (an explicit argument of the code, relative tolerance 10^{-12}). The informative below-branch ratio is 1.005 (LN) and 1.005 (SU) at $\sigma = 0.01$ and grows slowly with σ : the second order buys nothing at a crossing, in either branch. Away from the capacity ($\delta = -0.3$) the behaviour is that of the smooth frontier (minimum improvement ratio 282 for $\sigma \leq 0.05$) until the upper tail of the population reaches the capacity; at $\delta = -0.03$ the switch occurs inside the fitting window and gives the transitional slope 8.8 (0.11) of Figure 2. Units lie on both sides of the capacity in 107 of the 168 cells.

4.3 E3: two-level hierarchy

$G = 40$ firms of $n = 50$ units ($N = 2000$), firm means log-normal around the sector mean with between-firm dispersion σ_B , units log-normal around their firm mean with within-firm dispersion σ_W , both in $\{0.05, 0.1, 0.2, 0.4\}$, 20 replications; smooth Cobb–Douglas frontier and the same frontier capped at the sector mean output. Figure 2 (bottom); all cells, with bootstrap intervals, are in `results/tables.md`. The identity $\hat{Y}_2^{\text{TD}} = \hat{Y}_2$ of Lemma 3.6(i) holds to a relative 2×10^{-16} (floating point). On the smooth frontier the bottom-up estimator is better in 16 of 16 cells, by a median per-replication factor from 1.15 (small σ_B , large σ_W ; median errors

σ	no capacity						capacity +0%	
	rev. 1	rev. 2	rev. 3	cert. (micro)	cert. (mom.)	cert. rev.	rev. 1	rev. 2
0.02	0.2 [0.1, 0.7]	0.0 [0.0, 0.4]	0.0	100.0	100.0	0	31.2 [28.4, 34.1]	31.1 [28.3, 34.0]
0.05	0.6 [0.3, 1.3]	0.0 [0.0, 0.4]	0.0	99.7	99.6	0	35.3 [32.4, 38.3]	34.8 [31.9, 37.8]
0.1	1.3 [0.8, 2.2]	0.0 [0.0, 0.4]	0.0	97.0	94.4	0	45.8 [42.7, 48.9]	44.8 [41.7, 47.9]
0.2	4.9 [3.7, 6.4]	0.2 [0.1, 0.7]	0.1	70.9	6.5	0	67.8 [64.8, 70.6]	65.7 [62.7, 68.6]
0.4	18.1 [15.8, 20.6]	2.4 [1.6, 3.5]	4.4	0.0	0.0	0	70.0 [67.1, 72.8]	60.8 [57.7, 63.8]
0.8	46.1 [43.0, 49.2]	2.8 [1.9, 4.0]	90.8	0.0	0.0	0	64.0 [61.0, 66.9]	48.2 [45.1, 51.3]

Table 3: E4. Rank-reversal rates in percent at orders 1–3 over 1000 trials per cell, with Wilson 95% intervals for the rates that enter C2; “cert.” is the percentage of trials satisfying the margin condition with second-order bounds in micro-data and in moment-and-range form, and “cert. rev.” the number of reversals among the former (the latter are a subset). Order 3 with the capacity and the other capacity positions are in `results/tables.md`.

3.0×10^{-3} and 2.5×10^{-3}) to 4473 (large σ_B , small σ_W ; median errors 1.9×10^{-3} and 5.1×10^{-7}); the bottom-up error depends essentially on σ_W only, by Lemma 3.6(ii) and the homogeneity of the Cobb–Douglas frontier under multiplicative shocks ($f(\bar{x}_g \odot v) = f(\bar{x}_g)f(v)$, so a firm’s relative error does not depend on its mean), and the bound checks of Lemma 3.6(ii) and Proposition 3.1(c) pass in every replication (all hold: yes). With the capacity, 20–100% of the firms straddle it; the bottom-up error is 0.009–0.95 times the pooled one, and the firms that do not straddle contribute at most a fraction 6.9×10^{-3} of the bottom-up error: the hierarchy localises the failure but does not remove it, since inside a straddling firm Corollary 3.9 applies.

4.4 E4: rank reversals between two sectors

Two sectors of $N = 500$ units each. Sector A has mean $\bar{x}$ and dispersion σ ; sector B has mean $\bar{x}(1 + \gamma)^{1/0.8}$, so that $f(\bar{x}_B) = (1 + \gamma)f(\bar{x}_A)$ with γ uniform on $\pm 5\%$, and dispersion $\sigma/2$. Inputs log-normal; 1000 trials per cell; $\sigma \in \{0.02, \dots, 0.8\}$; capacity at $+50, +20, +10, +5, +0\%$ of $f(\bar{x}_A)$ or absent. A trial is a reversal at order k when $\text{sign}(\hat{Y}_k^A - \hat{Y}_k^B) \neq \text{sign}(Y^A - Y^B)$. In the smooth cells a comparison is *certified* when the margin condition of Proposition 3.12 holds with the bounds B_2 of the two samples, in micro-data and in moment-and-range form. Table 3, with Wilson 95% intervals; the other capacity positions are in `results/tables.md`.

Without capacity the second order removes almost all reversals: the first-order rate reaches 46.1% and the second-order rate never exceeds 2.8%. Among the certified comparisons there were 0 reversals at order 2 (and 0 among comparisons certified with first-order bounds), as Proposition 3.12 guarantees; but the micro-data certificate covers 100.0% of the trials at $\sigma = 0.02$, 97.0% at $\sigma = 0.1$, 70.9% at $\sigma = 0.2$ and 0.0% at $\sigma = 0.4$, and the moment-and-range certificate, the one available without the micro data, covers 100.0%, 94.4%, 6.5% and 0.0%. With the fourth-order bounds in the margin condition (Proposition 3.3(d)) the certified share at $\sigma = 0.2$ becomes 83.0% (micro data) and 11.5% (moments and range), and 95.7% and 71.6% with $S = \text{diag}(\bar{x})$ (third order: 90.0%, 56.7%); at $\sigma = 0.4$ (scaled, micro data) it is 1.2% against 3.4% at third order, so the smaller bound should be used; 0 of these certified comparisons are reversals. The third order is harmful at large dispersion: at $\sigma = 0.8$ its reversal rate is 90.8%, against 46.1% for the first and 2.8% for the second order, because the third central moments of log-normal inputs over-correct. With the capacity at the mean output of A the reversal rates are 31.2% (order 1) and 31.1% (order 2) already at $\sigma = 0.02$, and the second-order rate stays between 31.1% and 65.7%; with the capacity at $+10\%$ the second-order rate is 0.0%, 3.0% and 18.0% at $\sigma = 0.05, 0.1$ and 0.2 (first order: 0.6%, 4.3%, 23.1%), as the upper tail of A reaches the capacity.

Criterion C2. Of the 23 cells with a first-order reversal rate of at least 2% (3 without capacity, 20 of the 30 with it), 15 fail C2 on the point estimates, all with a capacity. With the Wilson intervals (a cell fails clearly when the lower limit of the second-order rate exceeds one fifth of the upper limit of the first-order rate, passes clearly in the mirror case, and is borderline otherwise), C2 fails clearly in 14 capacity cells, is borderline in 2 cells (no capacity, $\sigma = 0.4$; $+50\%$, $\sigma = 0.4$; in the smooth one the rates are 18.1% [15.8, 20.6] and 2.4% [1.6, 3.5]) and passes clearly in 2 smooth and 5 capacity cells. In the capacity cells that pass ($+50\%$, $\sigma = 0.2$; $+50\%$, $\sigma = 0.8$; $+20\%$, $\sigma = 0.8$; $+10\%$, $\sigma = 0.8$; $+5\%$, $\sigma = 0.8$) the sample mean of A stays below the capacity in at least 96% of the trials, so the second order is right for reasons unrelated to the crossing. C2 does not hold.

4.5 E5: a scalar recalibration

The record of the research line mentions a calibration that failed its criterion; the original procedure is not documented, so we test the simplest one. The second-order term is rescaled, $\hat{Y}_\lambda = \hat{Y}_1 + \lambda Q_2$, with λ fitted by least squares on a training design (Cobb–Douglas, LN, $N = 2000$, 10 replications, $\sigma \in \{0.05, 0.1, 0.2, 0.4\}$, four regimes: no capacity, and $f(\bar{x})/c - 1 \in \{-0.1, 0, +0.1\}$), either globally or separately by regime, and evaluated on new seeds at off-grid dispersions $\sigma \in \{0.03, 0.07, 0.15, 0.3, 0.6\}$ for both input laws; the criterion is C1 on

regime	log-normal (LN)			symmetric (SU)		
	order 2	global λ	regime λ	order 2	global λ	regime λ
smooth	2.22	0.44	3.39	19.18	0.80	6.50
below	1.37	0.80	0.47	1.33	0.80	0.47
at (all replications)	1.00	1.00	0.98	1.02	1.04	1.08
at (below branch only)	1.02	1.04	0.98	1.02	1.04	1.08
above	n/a	n/a	n/a	n/a	n/a	n/a

Table 4: E5. Minimum over the test dispersions and replications of $|E_1|/|Y - \hat{Y}_\lambda|$; C1 requires at least 5 everywhere. “At” is reported over all replications and over those whose sample mean falls below the capacity (LN: 20–30% above; SU: below branch by convention); “above” is n/a because $Q_2 \equiv 0$. Per-dispersion values in `results/tables.md`.

claim	status
First-order error is N times the Jensen gap, non-positive for concave f	classical
Remainder identities, bounds B_1, B_2 , improvement ratio $\geq 3\lambda/(M_3\rho)$, certificates $(k+1)B_2 < Q_2 $ (Prop. 3.1); bounds hold in every simulated population, $ E_2 /B_2 \leq 0.018$; certificate ($k = 1$, LN) up to $\sigma = 0.042$ from moments and range, 0.056 from micro data (E1)	classical (Taylor), assembled here; verified
Fourth-order certificate and margin with $ C_3 + B_3$ (Prop. 3.3), Cobb–Douglas suprema (Lem. 3.4); bounds hold in every simulated population; certificate ($k = 1$, LN) up to $\sigma = 0.075$ (moments and range), 0.100 (micro data); certified pairs at $\sigma = 0.2$: 83.0% / 11.5% (E1, E4); orders σ vs. σ^2 (Rem. 3.5)	[proved here; not yet checked by an independent referee]; orders heuristic; figures verified numerically
CES suprema by interval enclosure; CES certificate up to $\sigma = 0.032$ ($k = 1$, LN)	computational; rounding not directed
Exponents $2/4/4$ on symmetric inputs; for the second order on log-normal inputs between 3 and 4, from 3.16 ($N = 200$) to 3.96 ($N = 200000$) (Rem. 3.2, E1, E1c)	consequence of Prop. 3.1; verified
Third order does not improve on the second for log-normal inputs (Rem. 3.2, E1)	heuristic; verified
Top-down second order equals the pooled one (by construction of Def. 2.3); bottom-up error is the sum of within-firm errors, gain 1.15–4473 (Lem. 3.6, E3)	elementary; verified
Hinge identity (Lem. 3.7); $E_2(f_c) = -NT_u + O(\text{smooth terms})$ (Prop. 3.8); identity checked to 2×10^{-15} (E2)	elementary / proved here; verified
At a crossing the error is $\Theta(\sigma)$, $ E_1 / E_2 \rightarrow 1$, $T_u = \sigma\tau_z + O(\sigma^2)$ (Cor. 3.9); slope 1.00, $T_u/(\sigma\tau_z) = 0.997$ at $\sigma = 0.01$ (E2, SU)	proved here; verified
Populations with equal first three moments and aggregates differing at order σ (Ex. 3.11)	proved here
Margin condition preserves the ordering (Prop. 3.12); 0 reversals among certified pairs; certificate for 70.9% (micro data) and 6.5% (moments and range) of pairs at $\sigma = 0.2$ (E4)	elementary; verified
Covariance floor $O(\sigma/\sqrt{N})$ with independent efficiencies; dominates up to $\sigma = 0.18$ (Limitations, E1b)	elementary; verified
Predefined C1 (fivefold error reduction) holds on the SU sweep, fails on the LN sweep beyond $\sigma = 0.42$ and at crossings; a scalar recalibration of Q_2 does not meet it (E1, E2, E5). Predefined C2 (fivefold reduction of rank reversals) fails clearly in 14 of the 20 eligible capacity cells; 1 capacity cell and 1 cell without capacity borderline (E4)	verified
Statements about real sectors, estimated frontiers or policy	not addressed

Table 5: Claims and their status. “Verified” means reproduced by the script of Appendix B with the fixed seed; it is not a proof. The original content of the draft is the combination Prop. 3.8, Cor. 3.9, Ex. 3.11, the experiments and, since v0.4, Prop. 3.3 with Lem. 3.4 (not yet refereed).

the test grid. The fitted values are $\lambda = 2.25$ (global) and 0.89, 3.14, 4.72 for the smooth, below and at-capacity regimes; above capacity $Q_2 \equiv 0$, no λ exists and there is nothing to recalibrate, so that regime is reported as not applicable and excluded from the verdict. Table 4.

No variant meets C1 on the smooth, below and at-capacity regimes (order 2: no; global: no; by regime: no), but the failures have different causes. In the smooth regime the regime-specific factor meets C1 for $\sigma \leq 0.3$ (minimum ratio 9.23, LN) and fails only at $\sigma = 0.6$ (3.39), beyond the range where even the plain second order meets it (E1; here 2.22); the global factor is worse than the representative unit there (minimum ratio 0.44), because a factor fitted to crossing cells destroys the smooth-regime accuracy. Below capacity, at dispersions where no unit crosses, the ratio is $1/(\lambda - 1)$ by construction (0.80 global, 0.47 by regime), because regimes are defined by the position of the mean and not by whether units cross; where units do cross, the plain second order gains at most a factor 2.88 (LN), and the regime factor meets C1 at $\sigma \in \{0.15, 0.3\}$ (ratios 9.76, 6.47) but not at $\sigma = 0.6$ (1.53). The substantive failure is at the capacity: the below-branch ratios are 1.02, 1.04 and 0.98 (LN), so every variant is as good as the representative unit and no better, the mechanism of Example 3.11: the error to be removed scales as σ , the only available regressor as σ^2 , and no constant reconciles them across dispersions.

5 What can be claimed

Table 5 separates what is classical from what is proved here and from what is measured, and records that neither predefined criterion was met. The positive content is precise: in the smooth regime the second-order correction reduces the error by a factor of order at least $1/\sigma$ and comes with certificates, computable from the

moments and the input range, that remain conservative (on log-normal inputs, valid for $\sigma \leq 0.042$, and for $\sigma \leq 0.075$ in the fourth-order form, against an improvement observed up to $\sigma = 0.75$); at a capacity crossing it is no improvement at all, the missing quantity T_u is not a function of the moments, and no rescaling of the moment term supplies it.

Limitations. The frontier is known exactly; estimation error in f or in the moments is not modelled, and would add to every error reported. The derivative suprema have closed forms for Cobb–Douglas (Lemma 3.4); for CES they are enclosed by interval evaluation of the Faà di Bruno formula (Appendix B; within a factor 1.2 of a grid maximum on thin boxes, 30 on wide ones), which gives a first CES certificate ($k = 1$, LN, moments and range: $\sigma \leq 0.024$ at third order, 0.032 at fourth), with rounding covered by generous margins, not directed rounding: not a computer-assisted proof. The third-order bounds exceed the actual error by a factor of at least 56 in every replication and of up to 1064 in median, because they use absolute third moments where the actual error cancels; the fourth-order form removes that cause, and what remains is the norm inequality and the growth of the derivative suprema over the input range (Remark 3.5). With independent idiosyncratic efficiencies θ_i [5] the observed aggregate $\sum_i \theta_i f(x_i)$ differs from $\bar{\theta} \hat{Y}_k$ by $\bar{\theta} E_k$ plus $\sum_i (\theta_i - \bar{\theta})(f(x_i) - \bar{u})$, a covariance term of relative order $\sigma/\sqrt{N}$ that no moment correction removes; with $\theta_i \sim U(0.5, 1)$ it dominates the second-order remainder up to $\sigma = 0.18$ at $N = 2000$ (E1b, `results/tables.md`). Two input laws and two frontiers are a controlled design, not a survey of shapes; the crossing analysis covers the capacity kink $\min(f, c)$ and, by the same proof, any frontier with a first-derivative jump along a hypersurface, but not regime switches that also change the level. The hierarchy has two levels; deeper hierarchies compose Lemma 3.6. The decision experiment uses one margin distribution ($\pm 5\%$), one dispersion ratio between sectors and 1000 trials per cell, which leaves 2 of its 23 eligible cells undecided.

Next steps. (a) *Done in part* (v0.4): Proposition 3.3 exploits the cancellation (bound relative to $|Q_2|$ of order σ^2 instead of σ), but its reach is limited by the growth of $\|D^4 f\|$ over the coordinate range. Open: a region tighter than the coordinate box still known from moments (or a separate bound for the few extreme units); a fifth-order version with the signed fourth-moment tensor, which range inflation would hit harder; an optimised S (only I and $\text{diag}(\bar{x})$ were tried); CES enclosures with directed rounding. (b) The crossing term T_u is the Jensen gap of the hinge; estimating it from a parametric family fitted to $(\bar{x}, \Sigma)$ is exact within the family and should be tested out of family, with Example 3.11 as the worst case. (c) Efficiency terms correlated with inputs, where the covariance term acquires a mean that the moment expansion must model. (d) The dynamic element the line’s name promises: a trajectory σ_t or $\bar{x}_t$ approaching the capacity, for which Proposition 3.8 and Remark 3.10(i) already give $E_2(t)$ and the regime switch. (e) Any statement about a real sector needs an estimated frontier and its sampling error first; nothing here licenses that step.

A Proofs of Proposition 3.3 and Lemma 3.4

Proof of Proposition 3.3. As in Proposition 3.1, $\varphi_i(t) = f(\bar{x} + th_i)$ is C^4 on $[0, 1]$ with $\varphi_i^{(r)}(t) = D^r f(\bar{x} + th_i)[h_i, \dots, h_i]$, and Taylor’s formula with integral remainder gives $\varphi_i(1) = \sum_{r=0}^3 \varphi_i^{(r)}(0)/r! + \int_0^1 \frac{(1-t)^3}{3!} \varphi_i^{(4)}(t) dt$. Summing over i , the terms $r = 1$ cancel ($\sum_i h_i = 0$), the terms $r = 2$ give Q_2 and the terms $r = 3$ give $\frac{1}{6} \langle D^3 f(\bar{x}), \sum_i h_i^{\otimes 3} \rangle = C_3$; as $\hat{Y}_3 = Nf(\bar{x}) + Q_2 + C_3$, this is (a). (b): with $v_i = S^{-1}h_i$, $|D^4 f(y)[h_i, \dots, h_i]| = |D^4 f(y)[Sv_i, \dots, Sv_i]| \leq \|D^4 f(y)\|_S \|v_i\|^4$, $\int_0^1 \frac{(1-t)^3}{6} dt = \frac{1}{24}$, a convex R containing the population contains $\bar{x}$ and every segment $[\bar{x}, x_i]$, and $|E_2| \leq B_2$ is Proposition 3.1(c). (c): $E_1 = Q_2 + C_3 + E_3$, so $|E_1| \geq |Q_2 + C_3| - B_3 \geq |Q_2| - |C_3| - B_3 > k(|C_3| + B_3) \geq k|E_2|$. (d): $Y^A - Y^B = (\hat{Y}_3^A - \hat{Y}_3^B) + E_3^A - E_3^B$ by (a), and $|(\hat{Y}_3^A - \hat{Y}_3^B) - (\hat{Y}_2^A - \hat{Y}_2^B)| = |C_3^A - C_3^B| \leq |C_3^A| + |C_3^B|$, so the hypothesis gives $|\hat{Y}_3^A - \hat{Y}_3^B| > B_3^A + B_3^B$ with the sign of $\hat{Y}_2^A - \hat{Y}_2^B$, which then fixes the sign of $Y^A - Y^B$. $\square$

Proof of Lemma 3.4. $\partial^e f = A \prod_m \partial_m^{e_m} x_m^{a_m}$ and $\partial^n x^a = (a)_n x^{a-n}$; the exponent $a_m - e_m$ is positive if $e_m = 0$ and negative if $e_m \geq 1$ (as $a_m < 1$), which gives the monotonicity in each coordinate and the corner. Finally $D^k f(y)[Sv, \dots, Sv] = \sum_{j_1, \dots, j_k} s^e \partial^e f(y) v_{j_1} \cdots v_{j_k}$ is at most the Frobenius norm of the array $(s^e \partial^e f(y))$ times $\|v\|^k$ by Cauchy–Schwarz, and each entry is bounded by its value at its own corner. $\square$

B Reproducibility

The directory contains `experiments/frontier_aggregation.py` (E0–E6, flag `-fast` for a reduced run; one random generator per experiment spawned from the master seed) and `experiments/make_numbers.py`, which converts `results/results.json` into the macros and table bodies used by this document, so that every

measured number in the text is generated (the numbers typed in the text are design parameters and closed-form constants). The reference run used seed 20260930, Python 3.11.15, NumPy 2.4.6 [9] and SciPy 1.17.1 [18]; on shared cores it took 52 s of wall time for the computation and 56 s including tables and figures (53 and 57 s of CPU; interpreter start-up excluded), and its script hash is 5b309162009c. The closed-form gradient, Hessian and third-derivative tensor of both frontiers are implemented in the classes `CobbDouglas` and `CES` and checked by E0 against central differences. Bounds: for a Cobb–Douglas frontier $f(x) = A \prod_j x_j^{a_j}$ every entry of $D^k f$ has the form $\kappa A \prod_m x_m^{a_m - e_m}$, monotone in each coordinate, so its supremum over a coordinate box is attained at the corner given by the sign of each exponent, and the Frobenius norm of the entrywise suprema bounds $\|D^k f\|$ on the box. Micro-data (segmentwise) form: for each unit the box spanned by $\bar{x}$ and x_i , which contains the segment $[\bar{x}, x_i]$, gives $M_k^{(i)}$, and $B_1 = \frac{1}{2} \sum_i M_2^{(i)} \|h_i\|^2$, $B_2 = \frac{1}{6} \sum_i M_3^{(i)} \|h_i\|^3$. Moment-and-range form: one box given by the coordinate range of the population gives M_3 and $B_2 = \frac{N}{6} M_3 m_3$. Certificates: $(k+1)B_2 < |Q_2|$ in every replication, $k \in \{1, 5\}$, reported as the largest σ up to which they hold uniformly. Fourth-order certificate: `theory/check_sharp_certificate.py` reproduces the draws of E1 and E4, takes $D^4 f$ from Lemma 3.4 with the same corner rule (B_3 segmentwise or $\frac{N}{24} M_{4,S} m_4^S$ on the box; $S = I$ and $\text{diag}(\bar{x})$) and, for CES, the smaller of a naive interval enclosure and a centred form of each Faà di Bruno entry on a 4×4 subdivision of the box; its output `theory/sharp_certificate_results.json` (259 s of CPU; SHA-256 prefix a75d9e3fb136, script 8b6366176f05) is frozen in `results/sharp_certificate.sha256` and checked by `make_numbers.py`. E1c: 60 replications for each N , in blocks of 6. Input laws: correlated standard shocks z (Cholesky); SU uses $w = \sqrt{3}(2\Phi(z) - 1)$ on half the sample and $-w$ on the other half. E3: firm means $\bar{x} \odot \exp(\sigma_B \zeta - \sigma_B^2/2)$, units $\bar{x}_g \odot \exp(\sigma_W z - \sigma_W^2/2)$. E4: capacity $c = f(\bar{x}_A)(1 + p)$; Wilson intervals with $z = 1.96$. E5: $\lambda = \sum E_1 Q_2 / \sum Q_2^2$ over the training cells, globally or within each regime; evaluation on fresh shocks. Uncertainty: percentile bootstrap over replications (1000 resamples) for medians, and over replications with refitting (500 resamples) for exponents (E1, E1c, E2); one bootstrap generator per experiment.

References

- [1] Dennis Aigner, C. A. Knox Lovell, and Peter Schmidt. Formulation and estimation of stochastic frontier production function models. *Journal of Econometrics*, 6(1):21–37, 1977.
- [2] Kenneth J. Arrow, Hollis B. Chenery, Bagicha S. Minhas, and Robert M. Solow. Capital-labor substitution and economic efficiency. *Review of Economics and Statistics*, 43(3):225–250, 1961.
- [3] Charles W. Cobb and Paul H. Douglas. A theory of production. *American Economic Review*, 18(1, Supplement):139–165, 1928.
- [4] Jean Dieudonné. *Foundations of Modern Analysis*. Academic Press, New York, 1960.
- [5] Michael J. Farrell. The measurement of productive efficiency. *Journal of the Royal Statistical Society, Series A*, 120(3):253–290, 1957. doi: 10.2307/2343100.
- [6] Jesus Felipe and Franklin M. Fisher. Aggregation in production functions: what applied economists should know. *Metroeconomica*, 54(2–3):208–262, 2003.
- [7] Franklin M. Fisher. The existence of aggregate production functions. *Econometrica*, 37(4):553–577, 1969.
- [8] William M. Gorman. Community preference fields. *Econometrica*, 21(1):63–80, 1953.
- [9] Charles R. Harris, K. Jarrod Millman, Stéfán J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J. Smith, Robert Kern, Matti Picus, Stephan Hoyer, Marten H. van Kerkwijk, Matthew Brett, Allan Haldane, Jaime Fernández del Río, Mark Wiebe, Pearu Peterson, Pierre Gérard-Marchant, Kevin Sheppard, Tyler Reddy, Warren Weckesser, Hameer Abbasi, Christoph Gohlke, and Travis E. Oliphant. Array programming with NumPy. *Nature*, 585:357–362, 2020. doi: 10.1038/s41586-020-2649-2.
- [10] Hendrik S. Houthakker. The Pareto distribution and the Cobb–Douglas production function in activity analysis. *Review of Economic Studies*, 23(1):27–31, 1955.
- [11] Johan Ludwig William Valdemar Jensen. Sur les fonctions convexes et les inégalités entre les valeurs moyennes. *Acta Mathematica*, 30:175–193, 1906. doi: 10.1007/BF02418571.
- [12] Arthur Lewbel. Aggregation with log-linear models. *Review of Economic Studies*, 59(3):635–642, 1992.
- [13] André Nataf. Sur la possibilité de construction de certains macromodèles. *Econometrica*, 16(3):232–244, 1948.
- [14] Gary W. Oehlert. A note on the delta method. *The American Statistician*, 46(1):27–29, 1992.
- [15] Thomas M. Stoker. Completeness, distribution restrictions, and the form of aggregate functions. *Econometrica*, 52(4):887–907, 1984.
- [16] Henri Theil. *Linear Aggregation of Economic Relations*. North-Holland, Amsterdam, 1954.
- [17] Kees Jan van Garderen, Kevin Lee, and M. Hashem Pesaran. Cross-sectional aggregation of non-linear models. *Journal of Econometrics*, 95(2):285–331, 2000.
- [18] Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, David Cournapeau, Evgeni Burovski, Pearu Peterson, Warren Weckesser, Jonathan Bright, Stéfán J. van der Walt, Matthew Brett, Joshua Wilson, K. Jarrod Millman, Nikolay Mayorov, Andrew R. J. Nelson, Eric Jones, Robert Kern, Eric Larson, C J Carey, İlhan Polat, Yu Feng, Eric W. Moore, Jake VanderPlas, Denis Laxalde, Josef Perktold, Robert Cimrman, Ian Henriksen, E. A. Quintero, Charles R. Harris, Anne M. Archibald, Antônio H. Ribeiro, Fabian Pedregosa, Paul van Mulbregt, and SciPy 1.0 Contributors. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261–272, 2020. doi: 10.1038/s41592-019-0686-2.